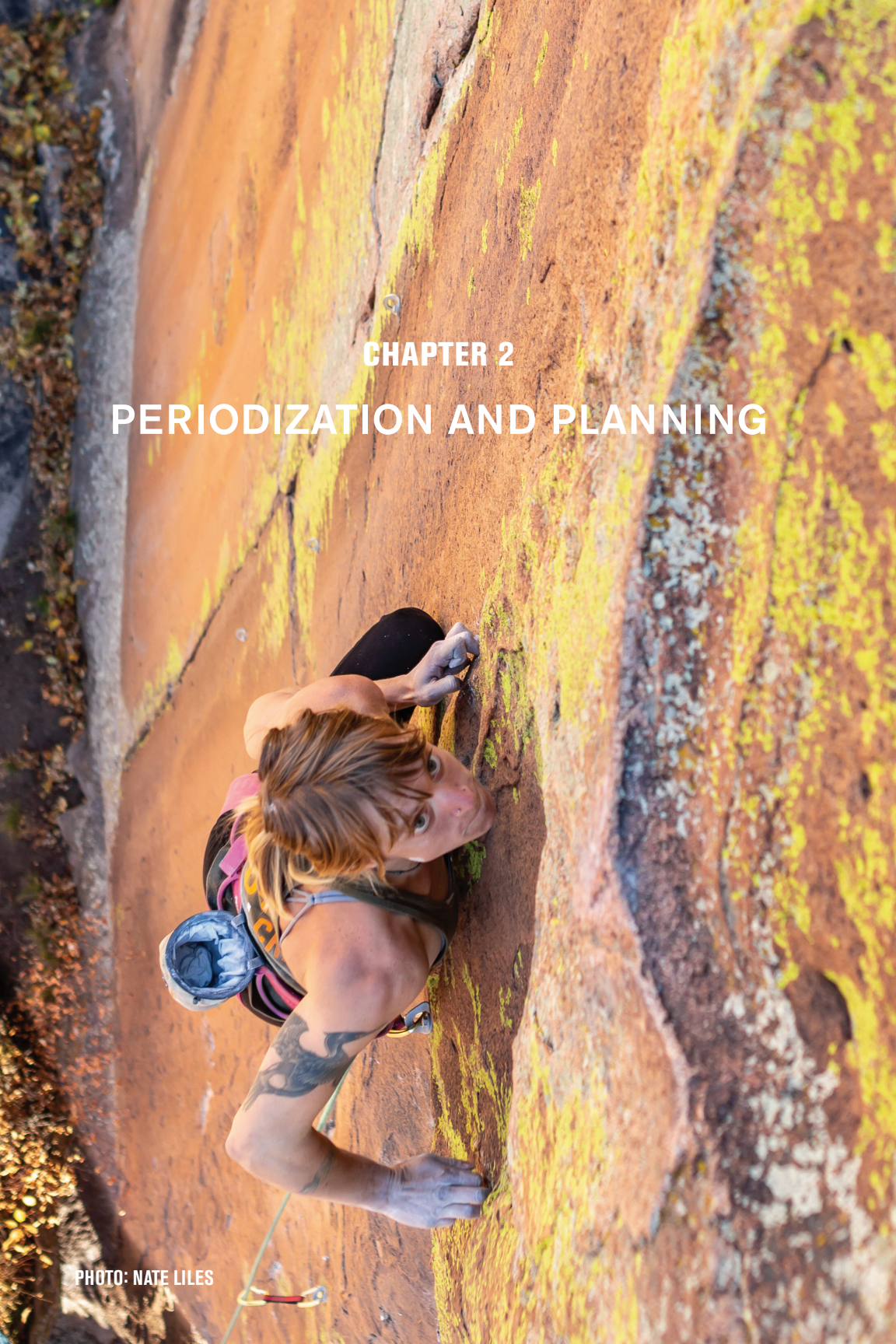


A female rock climber with blonde hair and a tattoo on her right arm is shown from the chest up, clinging to a reddish-orange rock face. She is wearing a blue and pink climbing harness and a blue helmet. The rock face is textured with yellow and orange hues, and a rope is visible in the background.

CHAPTER 2

PERIODIZATION AND PLANNING

PHOTO: NATE LILES

The older I get the less convinced I am of the superiority of any one training method. I used to be religious about structuring my season. I used to record my bodyweight before hangboard workouts so as to adjust the load specifically with the half-pound weights I bought for the job. I would hardline it on the phases, flat out refusing to climb cruxy routes of any description during an endurance phase.

Eventually I started paying attention to what other people were doing in practice, not just what should happen in theory. I also started looking at how closely I truly adhered to my training. Both of these activities were enlightening. The more people I talked to, the more I understood a couple of things. First: climbing is more interesting than training and this is probably a clue as to why we all picked the sport to begin with. Second: these people are not dumb. Most of the climbers that I end up talking to on the phone have read several books on climbing training and training in general, and still do not adhere to those plans exactly.

Despite this, I was talking to high performers. So instead of looking for why they didn't follow the recommended programs or looking for the flaws in their training (based on my vast book-based knowledge on how I thought they *should* be training), I began to look for what was *right* about what they were doing. Most of them started easy and made things harder over a few months' time. Most of them were aimed at one or two particular goals. Most had one training focus leading up to performance time. All of them climbed a lot.

At the same time, I compared my training log to my plan. This is a fascinating thing to do. It's easy for me to lay out a fantastic-

looking spreadsheet, but the day-to-day gets much harder. As Morpheus said in the Matrix, “There is a difference between knowing the path and walking the path.” I was lucky to hit 50% of my planned training. Whether it was bad coaching or a bad athlete, I don’t know...but something had to be done.

All of these things helped me to question my assumptions... which sucks. My assumptions help keep my ego warm.

The problem with learning from all of these climbers, of course, is that no one had the same plan. In fact, no one seemed to have the same plan season to season themselves! I decided to start by filtering out any common traits and starting there. The last piece of the puzzle was figuring out just what kind of activity rock climbing really was. See, most of us read a book on running or weight lifting and can tease out parallel tracks. We can imagine how this or that technique might apply to our sport, how if runners build endurance using “x” mode, we could morph it to climbing. We then make a fatal jump in assuming climbing is exactly like said sport.

Weightlifting is a strength sport. Five seconds (or less) and you rest. Running and cycling and swimming are all “cyclic aerobic sports”, which are very energy-system based. Climbing is neither a strength sport nor a cyclic endurance activity. Climbing is a multi-element skill sport that requires a high level of strength, explosiveness, endurance, and mobility. This causes a couple of problems. One, you have to train and perform in multiple energy systems. This means you might need explosive strength after fifteen minutes of easy aerobic activity. It means you might depend on the lactic energy system heavily for one pitch, followed by primarily alactic energy the next.

The second problem is skill. Climbing is all about execution of highly coordinated skills, and your skill is simply supported by your strength and conditioning. This means that not only do you have to train across multiple energy systems, but you have to do it “behind” your technique training. This is one complicated mess.

My original periodized plans clearly didn't work. From the ages of 16 to 18 my climbing improved by one number grade per year, then hit the brakes big time. Later, during college and for a couple of years thereafter, I kept getting into and out of shape for the same grade. In my thirties, I crept another letter grade, but still only hit one or two hard redpoints a year. Structured training or not, I probably would have had about the same result. I knew this wasn't as good as it could get, but no one could help me find out why.

Eventually, I started looking for programs that would provide longer-lasting adaptations, with the end goal of keeping strong through a whole cycle. I also started looking for a program that I might follow through with. This search led me to the original writings on periodization from the Eastern Bloc coaches, and to the world of training for other skill-based performers from gymnasts to wrestlers to golfers...and even to musicians.

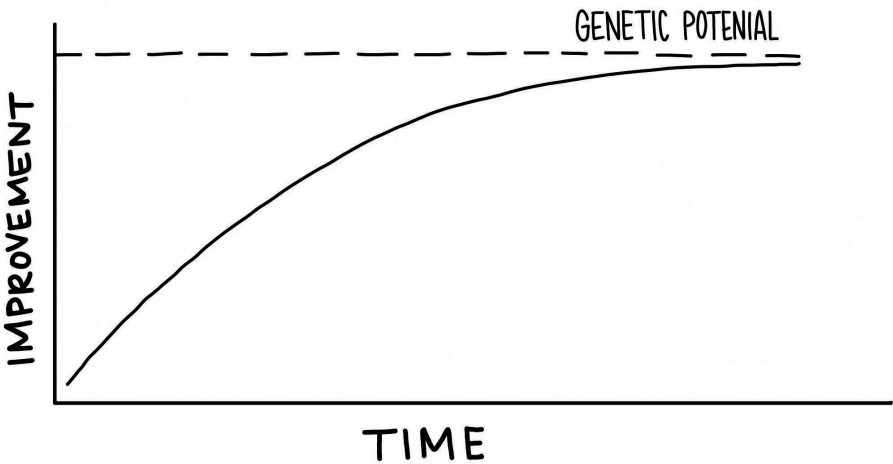
I learned about block periodization, where you train all qualities at all times, but focus on only one or two during any given phase. I learned about alternating linear training where one simply switched between two major foci for a protracted period, such as a month of strength followed by a month of power followed by a month of strength, and so on. I also studied what's called nonlinear periodization where one switches training foci on a weekly or session-to-session basis. I believe that for a sport such as climbing, these "non-traditional" types of planning are more appropriate than the traditional modes of classical periodization.

I am not all that convinced that there is any program that is fundamentally "better" than any other. It might be as simple as hitting your strength and power a couple of days a week and slowly ramp up your redpoint level over a couple of months. I have realized that any program that produces improvements in your climbing is probably a good one, and you should stick with it as long as it keeps paying out. Conversely, if you are staying

at the same level season after season, there is clearly something wrong. If there is anything that we are learning about our sport as it develops, it's that you can get better and better for a long time.

THE WHY AND HOW OF PROGRAM DESIGN

Based on what you read above, you might think that just throwing in the towel and forgetting about program design might be a good idea. For a first-time climber, that is probably not a bad choice. Yet as we develop fitness qualities, our rate of improvement declines. Think about how it felt to figure out how to hold slopers, how to drop-knee, or how it was when you learned how to de-pump on a route. These are huge advances in your climbing, and they probably happened in the first year or two of your climbing career.



Over time, we all reach a place along the improvement curve where progress slows so much it seems to have stopped. At this point many climbers simply cycle between being “in shape” and “out of shape” - sometimes for years. This is the point at which structured training becomes important. But don’t get ahead of yourself.

Structure can be very simple. Progress can be regained simply by assuring you increase the difficulty of your training for a few weeks. You can see huge gains in finger strength by simply hangboarding, at all, regardless of the specifics of the workout, a couple of times a week. This is where everyone should start. Sure, the promise of a confining year-long program that has produced V14 climbers seems attractive, but your program need not be that restrictive, yet.

We can categorize climbers based on training age. These categories are independent of ability, but are based on how much actual training the climber has done. The simplest plans work very well for the first couple of training years, and we call these Level I plans. The Level II plans are more involved, and feature smaller steps in progress. Finally, the Level III plans are restrictive and produce small results over long cycles. The Level III plans are appropriate only for athletes who have trained for several years and have ceased to progress.

It is not uncommon for an experienced climber to see huge benefits from the simple Level I programs, and the vast majority of veteran climbers are happiest with the Level II programs. It is critical to understand that the higher level, more complex programs are not somehow better - they are more focused on the minutiae. These programs are restrictive and produce results slowly. These programs aim to improve small facets of performance, and are thus *inferior* for the majority of climbers. I encourage all readers to start with shorter and simpler programs and work up the chain as progress slows. I coach several V-double digit and 5.14 climbers who never leave the Level I plans.

In the second edition of this book, I introduce the idea of adding a second level of planning on top of the nonlinear plans we discussed the first time around. This is called **Block Programming**. These blocks constitute a better way of advancing training once a simple nonlinear cycle plays out. Instead of simply aiming at switching between strength and power and endurance, a block program might have you doing a *strength phase* consisting of two strength sessions in a row, followed by a “maintenance session” of power and endurance. For a full 4-6 weeks, you would follow this cycle before adjusting to another focus. You’d look for big gains in strength and only aim to maintain everything else. More on this in a few pages.

Go as simple as possible, always: completing a simple 4-week plan is better than half-assing a complex 12-week cycle, and you’ll be happier with the results.

PERIODIZATION

At first, the training principles outlined above are pretty easy to follow. First-time athletes and young adults can often reap big benefits from a very basic overload program. Such a program would maintain roughly the same schedule week-to-week and might produce results for six to eight months without changing the schedule. The longer you stick to a program, the more you’ll see your performance start to level off. This is where the entire concept of periodized training comes in. Periodization, in its simplest form is just changing the primary type of training you are doing from time to time. We naturally tend toward some degree of periodization - whether it’s simple “on” and “off” seasons, or a complex week-to-week detailed calendar. Any athlete that performs at a high

level will notice ebbs and flows in their abilities and motivation. Periodization is simply planning to make the most of these ebbs and flows.

Knowing that most athletes can only adapt to a given training stimulus for 6 to 8 weeks before levelling out, coaches have successfully developed program cycles to prevent plateauing. Such programs are all designed to improve athleticism and performance by:

- Limiting decline of skills between training cycles
- Allowing the athlete to adapt to one skill at a time
- Controlling the inevitable build-up of fatigue that comes with progressive training

The history of periodized training goes back to the dawn of athletics in ancient Greece, when there were times of *hardening* for events and times of rest. In the hundreds of years that followed, athletes and soldiers the world over followed similar cycles. It wasn't until the last century that anyone really passed much detailed information down.

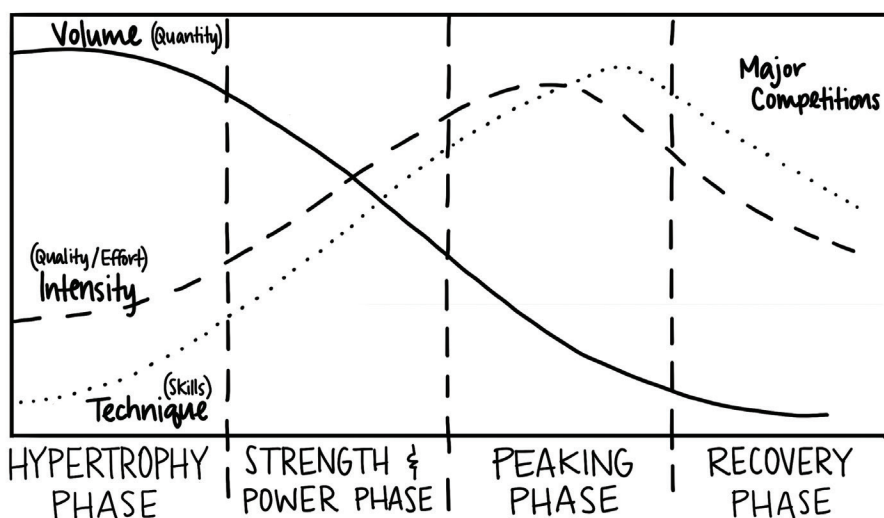
The recent history of periodization is an interesting one, and was heavily dependent on people stealing each others' ideas. A short version, from Practical Programming by Mark Rippetoe and Lon Kilgore is this:

“As early as 1933, Mark Berry was using weekly periodization for his bodybuilders and weightlifters, and wrote about it in several publications. In the 1950s, Lazlo Nadori, a sports scientist and coach in Hungary, developed a model of periodization for his athletes. The development of this particular model was unique to Hungary and was separate from the evolution of Soviet Bloc periodization, since no translations were done. In the 1960s, Russian weightlifting coach Leonid

Matveyev developed his concepts of periodization, and his 1971 book provides several different models. The one of these that is now known as “classical” periodization was intended for beginners. Later in the 70s, Matveyev’s book was translated into German and English. Having written the first periodization book in the West, he became known as the father of periodization by default.

*Also in the 1960s, Matveyev’s hated rival, Yuri Verkhoshansky developed his system of conjugated loading, openly stating that periodization was crap. But since his conjugated loading system was also periodized, it must be assumed that that he really just thought Matveyev’s approach to periodization was crap. In 1982, an east German sports scientist, Dr. Dietrich Harre, edited principles of sports training, which is essentially a fusion of the works of Nadori and Matveyev. A couple of years later, Frank Dick, head of British track and field, “liberally recreated” Harre’s book in English. Tudor Bompa, author of the famous text *Periodization*, was trained in the East German system, and his first and subsequent texts are essentially reiterations and adaptations of Harre’s adaptation of Nadori and Matveyev. And here we are today with no new thoughts, no new systems, and no real explanations of periodization since the last century. What we do have is a large misunderstanding of periodization and how to use it.”*

For those unfamiliar with the basic structure of classical periodization, here is the ten-second version: Training is broken up into several *mesocycles*, which are typically about a month long. The athlete focuses on one fitness quality for this cycle, progressing it as far as possible before switching to the next cycle. Traditional modes either progress from long to short or from short to long.



What this means is you move from one end of the volume/intensity spectrum to the other as the cycle progresses. The most popular linear climbing cycles presently are built short to long. We do short, high intensity sessions early in the phase, followed by sessions that are slightly lower intensity, but with higher durations, then finally we move to relatively low intensity, high volume work as we start to redpoint. Others argue the opposite approach, and such an approach has been used successfully by top redpoint climbers and boulderers for years. In a long to short plan, you simply start with high volume training and add intensity as the cycles progress, ending the training phase with focused maximum strength work and lower total training volume.

THE FLAWS OF LINEAR PLANNING IN SKILL SPORTS

I have now read no fewer than twelve books by eight different experts that were supposed to have “invented” periodized training. It doesn’t matter who the father of modern periodization is; the whole idea - cycling training according to a sports season



PHOTO: JOSH LARSON

to enhance results in performance - is great. The problem is that it's not ideal for rock climbing.

I'll be the first to admit it's attractive...the smooth transition between foci, planning for peak performances during certain times of the year, the forced periods of layoff. Unfortunately, there are four big problems with such planning.

First, the whole model was built around peaking for power sports which have a clear and very limited competition schedule. Athletes needed to be very fit for a competition, and such planning worked pretty well to get them there. But with over 50 years of programming, coaches are still unable to accurately plan for personal best performances. Yes, we can get you pretty fit, but no, we can't guarantee you'll be the best you can be. The problem is magnified when your sport requires multiple periods of high performance...like we hope for in climbing.

A sad commentary on planning is shown by the fact that only about 20% of Olympic athletes turn in their best performances at the games. Think about it: one out of 5 get it right for the single most important event of their athletic career.

Second, rock climbing is not exactly an "acyclic power sport," as Matveyev describes in his original text (the one upon which we built that whole idea of classical progressions). It is a skill sport, supported by strength and endurance. Skills must be constantly and continuously trained and developed, using a high level of volume throughout the year. In sports like climbing, physical readiness is not even half of the story. This works out well for climbers, though. The protracted periods required to develop skill allow us to extend the number of great climbing days each season, provided we build the program correctly. If it were all physical readiness for a lower-skill movement, fitness peaks would matter greatly and the loss of even a little strength or endurance would be painfully clear.

Third, biological organisms don't adapt according to time. The number of *discrete overload events* (workouts) is the key. Trying to

force a strength adaptation in 4 weeks, followed by power, then by power endurance might very well work for some, but what if you adapt more slowly? What if you have a bad week, or a bad month? What if you're 40? 50? A too-rigid plan leads almost inevitably to failure. The more flexible your planning, the more likely you are to hit performance peaks.

Finally, and most importantly, periodized plans slowly cycle through the energy systems as the plan progresses, all with the goal of elevating a single facet of fitness at the end of several months, such as power. This works pretty well, but in climbing we've got a problem: the sport requires high levels of strength, of endurance, of anaerobic capacity, and power. It is hugely challenging to get this done over a prolonged season with long training cycles. Because of the varying levels of *adaptation persistence* of the various facets of training, we cannot simply stop training any facet of the sport for very long.

The continuity of the training program over time is critical. The lack of this continuity is the primary argument that critics of periodized planning put forth. Charlie Francis, the greatest track coach of all time, found that a typical plan for sprinters called for many weeks at a time of base training or overdistance training - which he felt was detrimental to his runners' speed. At the same time, the ramp-up to the sprinting phase was too steep and would create an injury risk. He found that running at even 85% of maximum pace was a totally different *skill* than maximum pace and yet this is how many coaches trained their athletes to prepare for the season (and still do). Instead, he taught that one should train at high speed/effort throughout the year, but vary the volumes of this intense kind of training. This is exactly what you are going to do.

Yes, we hit plateaus in training after 4-6 weeks, but completely changing your training focus is the easy way out. It tricks you into thinking you're making progress because of the novelty of the new

program, which can leave you tired and sore during the first weeks of change. Novelty, not progress.

A typical rock climbing program might suggest the same: A month of hangboard, a month on the campus board, a few weeks of 4x4s...which might mean three months away from high-level climbing, and many weeks away from critical skill-building stimuli. And once you're climbing, where does that hard-earned strength go? Imagine a golfer, a figure skater, a trials rider...spending three months partitioning their training in such a way that they don't actually do their sport...these athletes would never perform at their potential. We simply cannot ever stop working on movement, on strength, or on endurance.

ADAPTATION PERSISTENCE

All this leads us to the idea of *adaptation persistence*, which is simply how long a given facet of training takes to build and how long it takes to go away. Certain qualities develop fairly quickly, such as cardiovascular endurance. Others, such as muscular hypertrophy or strength take a long time to develop. This is true on the downhill end, too. Each of the primary fitness qualities detrains similarly to how long it takes to train. Hypertrophy, once gained, sticks around a long time. Cardiovascular endurance, as endurance athletes know, begins detraining in as little as 24 hours...thus the need to run or ride or ski almost daily to maintain fitness.

Add to this the fact that climbing requires several skills to be developed at once to assure high performance, and you can see why letting any of the facets degrade too far can result in poor performance. There have been many attempts at rectifying performance decay in training, but most of them have common features. In 2014, Greg Nickols and Mike Israetel wrote a great article called *There Is Only One Type of Periodization*. In it, they

explain that all programming comes down to manipulating three variables:

1. **Linearity.** Your program can progress in a linear fashion, simply overloading the same system over and over, either with increases in volume or load or both. You can also choose not to do this, but rather progress multiple factors at the same time. All programs that are not linear, are (surprise!) called nonlinear. These programs can change loading parameters on a set-by-set scale, a workout-by-workout scale, or even a week-by-week scale.
2. **Undulation.** Related to linearity, undulating your training involves changing loads within workouts or weeks. By subjecting the athlete to contrasts in load, the system continually adapts to several levels of intensity.
3. **Conjugation.** This is the regular changing of training stressors, either by type of load (strength versus explosiveness) or type of movement (campusing versus bouldering). Again, this factor can change on many scales, from daily to monthly.

This allows that several training programs might work well for any given climber, and the plan should be chosen carefully. I can't emphasize enough the critical aspect of choosing your program based on what you are doing, or have done, instead of what you hope you could do. It is infinitely better to run through a 4-week basic program and add complexity to it than to start a 16-week advanced plan and try to dumb it down.

No program is appropriate all the time for any athlete. Most accomplished climbers have tried several programs, taken the gems from them and created a personalized program that works for them. This should be the end-goal for every athlete. As we build our planning, we need to keep in mind which factors are the ones we want to develop, which ones to maintain, and most

importantly, how little we can train in order to maintain. I'll touch on this more in a bit.

A HIERARCHY OF TRAINING MODES

If you were to look at the various training modes suggested over the years, you'd see a simple hierarchy start to appear. The programs at the bottom of the pyramid would be the simplest and most effective, with each subsequent level getting more complex, more restrictive, and less effective. The simple reality of this sequence is that the better you get, the harder it is to get better. As always, start with the simplest intervention first.

HIERARCHY OF TRAINING MODES

BLOCK PROGRAMMING

NONLINEAR PERIODIZATION

ALTERNATING-LINEAR CYCLES

LINEAR PROGRESSION

At the very bottom would be simple linear progression. You'd have a fixed workout week, and slowly progress the load in each workout as you got better. As I said above, a trainee might advance for several months on such a program. This type of progression would work well for any athlete adding in new exercises or workout styles anytime in their careers.

The next layer up would be what's called an alternating linear cycle. You'd do a simple linear progression for 3-4 weeks, then switch to another linear progression that focuses on different training factors. This surprisingly small change in programming could be used for years without a loss in effectiveness. I think switching between a power/strength focus for 3 weeks and a capacity focus for 3 weeks is a home-run program, though a bit too simple for most people to believe in. Simplifying one's program takes a rare courage few of us possess.

Further up the pyramid, we'd move into cyclic programs. These are the nonlinear programs, and would feature switching between session types on a daily or weekly basis. It is my belief that the majority of active climbers in the world would see the most progress on this type of plan, and this is the basis for the nonlinear training plans in this book.

At the very advanced level, we have the most restrictive plans, which are block programs. This type of plan asks that you dedicate most of your time in any given cycle to improving one facet of fitness while continuing to maintain the others with the bare minimum of work. These are plans appropriate for very advanced trainees, and I will detail this idea later.

In the end, we understand that the best training programs are the ones you'll follow to completion. Hope in your abilities to get disciplined or motivated does not lead to good training. Kicking the hell out of a simple plan is a much better bet.

NONLINEAR PROGRAMMING (PROS AND CONS)

After years of not getting what I wanted out of linear programs, I started searching out other programming structures. I had four criteria in mind that governed how I looked all the programs:

1. The program had to be simple to implement on a normal athlete's schedule. The vast majority of climbers I consulted were 9-to-5 workers who trained on a restricted schedule and climbed on weekends.
2. The program had to allow for longer periods of performance than traditional models. Linear programs bring you to a high peak of fitness a couple of times a year. I wanted a program that got a climber to a sustainable level of fitness that would last for several weeks at a time, several times a year. By necessity, such a program would not result in as high a peak in conditioning, but this is not a problem since climbing is only partly dependent on your conditioning... what really matters is being able to be strong enough to execute skill maximally.
3. The program had to be flexible enough that a week off here or a slight injury there didn't derail the whole season.
4. The program had to develop all of the relevant fitness qualities needed for climbing, and keep them from degrading as the season progresses.

There were many programs that fit most of the criteria above, but the best fit was a nonlinear cycle of training. What I liked most about the nonlinear cycles were their flexibility and the fact that

the research behind them was solid. As I said above, I didn't really care if we sacrificed peak fitness for a longer season or a more flexible schedule, but it turns out, there may be little to sacrifice. Remember, fitness is only one factor in top climbing performance.

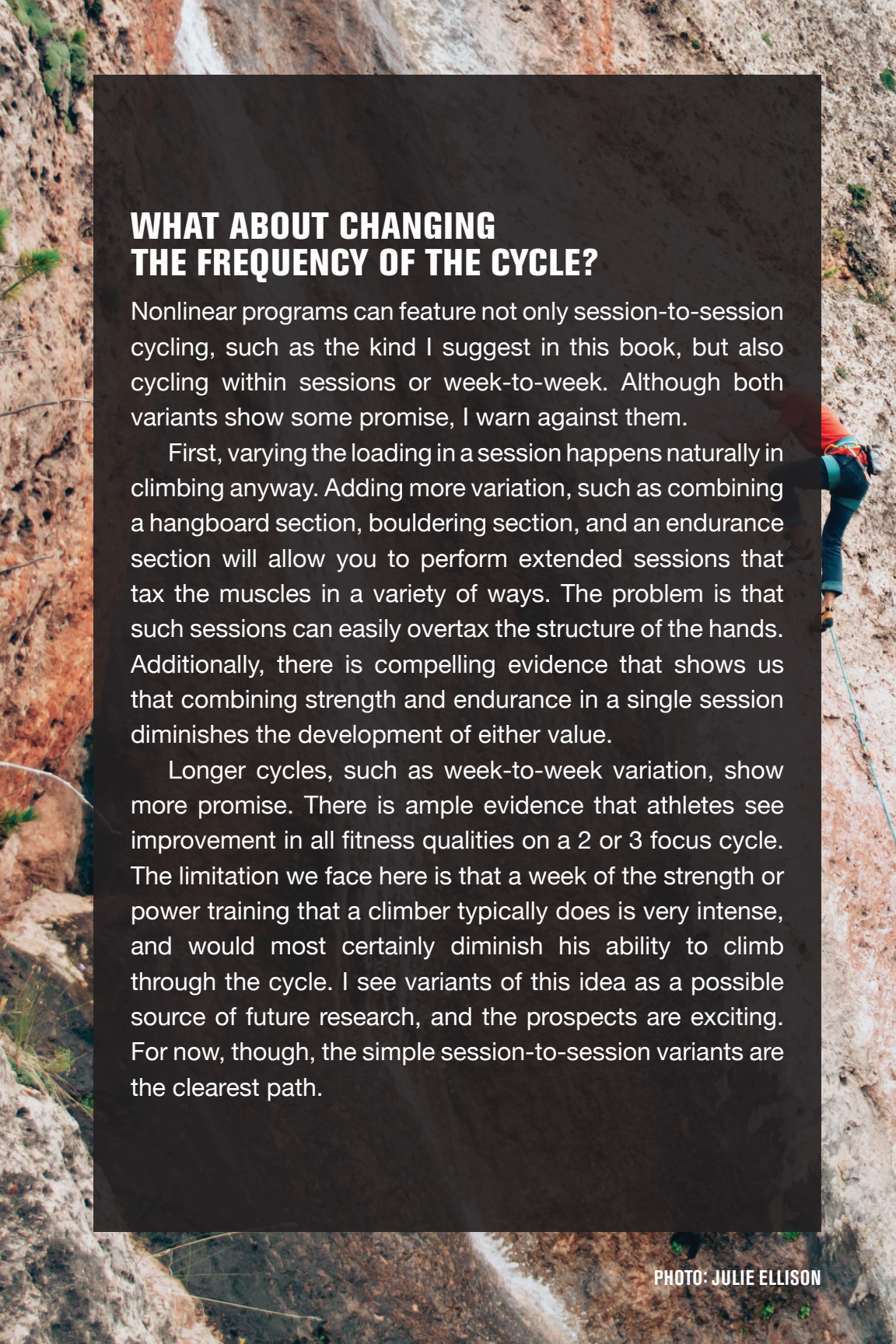
As Fleck and Kraemer put it in *Optimizing Strength Training*, “Nonlinear periodization is more flexible in how and when a peak in performance is created, depending on the goals of that particular mesocycle. It also allows for more frequent exposure to different loading stimuli within a weekly workout profile.” They go on to note that several studies compare this type of program favorably against a traditional model in overall gains in strength and fitness.

Of particular interest to climbers is Silvestre's (et al 2006) study where it was shown that competitive soccer players could maintain or even increase strength *in-season* on a nonlinear resistance training plan. Rhea (et al 2002) showed that session-by-session changes in training loads could increase pure strength by greater than 50% more than a traditional program.

Artur Montiero and colleagues compared nonlinear, linear, and nonperiodized modes of training in a group of experienced strength athletes. That the linear and nonlinear programs outperformed the nonperiodized group was of no surprise. What was interesting is that the nonlinear group showed strength gains throughout the 12 week cycle (testing at weeks 4, 8, and 12), where the linear group only showed gains at the end of week 8. The overall strength gains of the nonlinear group were significantly greater overall.

Continual gains in strength throughout nonlinear cycles have been shown time and again in research, including Kraemer's landmark study (2000) of female Division I tennis players, and the 2001 study by Marx, et al.

Not as big an issue for climbers, but a significant finding nonetheless: many of these studies clearly demonstrated greater changes in body composition or maintenance of ideal bodyweight in the participants. For many climbers I have consulted, the

A person in a red shirt and blue pants is climbing a rock face. A waterfall is visible in the background. The text is overlaid on a dark, semi-transparent rectangle.

WHAT ABOUT CHANGING THE FREQUENCY OF THE CYCLE?

Nonlinear programs can feature not only session-to-session cycling, such as the kind I suggest in this book, but also cycling within sessions or week-to-week. Although both variants show some promise, I warn against them.

First, varying the loading in a session happens naturally in climbing anyway. Adding more variation, such as combining a hangboard section, bouldering section, and an endurance section will allow you to perform extended sessions that tax the muscles in a variety of ways. The problem is that such sessions can easily overtax the structure of the hands. Additionally, there is compelling evidence that shows us that combining strength and endurance in a single session diminishes the development of either value.

Longer cycles, such as week-to-week variation, show more promise. There is ample evidence that athletes see improvement in all fitness qualities on a 2 or 3 focus cycle. The limitation we face here is that a week of the strength or power training that a climber typically does is very intense, and would most certainly diminish his ability to climb through the cycle. I see variants of this idea as a possible source of future research, and the prospects are exciting. For now, though, the simple session-to-session variants are the clearest path.

bodyweight gain in off-season has become a real struggle, and is one that does not diminish with age. Add this to the great training results seen in nonlinear plans and you have a winning program.

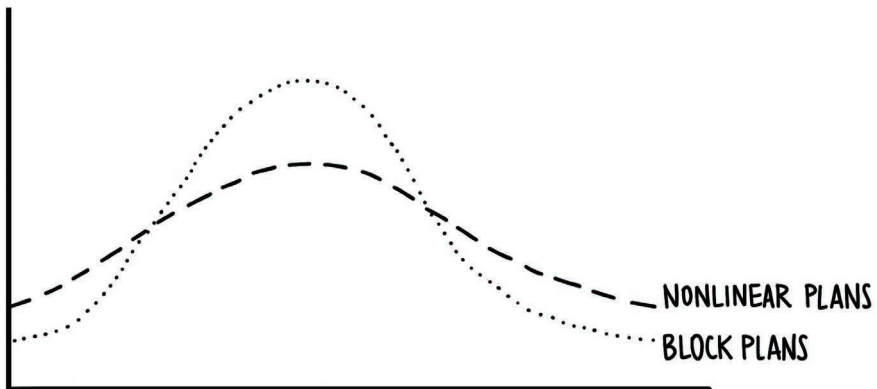
I have seen similar good results with climbers. Additionally, climbers find the nonlinear program more appealing for the same few reasons:

1. The program is “less boring” than one in which several sessions in a row are similar or the same.
2. It allows the climbers to more easily start up again after illness or injury.
3. It allows for more variety in workout sequence and in the facility used.
4. The program is adaptable to various schedule constraints such as limited workout time or crowded gym times.
5. Climbers get longer rest periods for certain tissues and muscles due to the varied loading throughout the week.

The big leap for me was not so much learning about this kind of cycling, but committing to it for a whole training cycle. Abandoning the training I was doing, no matter how mediocre its results, was hard. I encourage you to take a chance on trying something new, and pay attention to what happens.

BLOCK PROGRAMMING (PROS AND CONS)

As with any programming template, the block program has a lot to offer, and it has some drawbacks, too. The biggest drawback with the programming goes hand-in-hand with being a highly developed athlete: the closer you are to your potential, the harder it is to stay in top condition. In the nonlinear programs, we have great success holding a high level of fitness for long periods of time. In block programming, we are essentially looking for even higher levels of performance, and it comes with the pricetag of shorter phases of performance at max levels.



Discipline and long-term thinking are a must. If you tend to get bored with a program or are expecting quick results, a necessarily long-term program like a block plan is not for you. The only way this kind of plan produces superior results is in working your way through months of careful and progressed training. If you need a quick fix, the nonlinear plans are a better fit.

The primary benefit of block programming is that you don't have to have overly-long preparatory periods. You are spending a fair amount of training time maintaining several facets of your climbing fitness, so you don't have to go back and rebuild them each season. With the nonlinear / block hybrid I suggest in this book - essentially having differently-targeted monthly-long blocks of nonlinear cycles - you will maintain the flexibility that comes with the original Logical Progression plans, but can do a better job of targeting fitness peaks.

UNDERSTANDING NOTATION

ADD 25% VOLUME TO ARC SESSIONS, CAN BE AT CRAG OR GYM	HB LADDERS 2X PER WEEK, 3 POS, 5 SETS, SAME LOAD AS PREVIOUS WEEK	BOULDER INCRE
EXT-E 2X, 3X10@2, 3X12@2	HB LADDERS 2X PER WEEK, 3 POS, 5 SETS, INCREASE LOADS	BOULDER INCREASE
EXT-E 2X, 3X14@2, 3X16@2	HB LADDERS 1X, 3 POS, 6 SETS, INCREASE LOADS / HB REPEATERS, 4 POS, 2 SETS OF 6X 5/5	BOU
EXT-E 2X, 3X18@2, 4X9@1:30	HB LADDERS 1X, 3 POS, 6 SETS, MAINTAIN LOADS / HB REPEATERS, 5 POS, 2 SETS OF 6X 5/5	BOULDER 2
EXT-E 2X, 4X10:30@1:30, 4X12@1:30	HB LADDERS 1X, 3 POS, 6 SETS, MAINTAIN LOADS / HB REPEATERS, 5 POS, 2 SETS OF 5/5, INCREASE LOADS	BOULDER 2X40
EXT-E 1X, 3X12@1:30	HB LADDERS 1X, 3 POS, 6 SETS, MAINTAIN LOADS / HB REPEATERS, 5 POS, 2 SETS OF 5/5, INCREASE LOADS IF POSSIBLE	BOULDER 2X45

Over the past few years, a large portion of the questions we receive at Climb Strong about our books have to do with simple notation. The error is on our end - assuming someone understands how to

read a workout or a plan is the worst first mistake. If they can't even read the thing, how can we expect them to do it?

This notation is important. It is something that is easy to understand once you have the hang of it, and it opens up a valuable communication channel for athletes and coaches once both are on the same page. What follows are the most common terms, phrases, and notations you'll see in our training plans and articles. These terms are used consistently in our work and although you may see some variation from other coaches or cultures, with a good grasp of the basics you should be able to figure things out soon enough.

EXERCISES

This might seem obvious, but an exercise is a specific pattern of movements designed to overload the athlete in a way that improves her function either as a human or as an athlete. Exercises are grouped together to form training sessions. Examples of exercises might be a squat, an edge hang, or a campus ladder.

SESSIONS

Sessions are groups of exercises. Also called workouts, sessions are primarily aimed at overloading a specific kind of fitness such as strength or endurance. Sessions are ideally grouped into a series that would then form a Training Plan.

TRAINING PLANS

These are groups of sessions aimed at eliciting a specific response from an athlete. This is a fundamental building block of progression, but is sometimes ignored, even by very good athletes. A training plan is generally 4-6 weeks in length and will focus on improving only one or two fitness qualities at a time.

PROGRAMS

Programs are groups of training plans. Most programs are 6 months to 5 years in length, and are built around long-term performance goals. Programs are normally quite detailed in the near-future and become less-so the further we look down the road.

SETS

These are individual cycles of exercises within a session. Most exercises are repeated several times in a session. For example, you might do 3 sets of squats and 3 sets of pull-ups in a given session. This means you would visit each exercise 3 different times in one session. Sets are divided into reps, which are the individual performances of the exercise.

REPS (OR REPETITIONS)

Reps are the number of times you perform an exercise in a given set. A session might call for 3 sets of 10 repetitions of a pull-up. This would mean you'll perform 10 pull-ups, then rest until recovered enough to do pull-ups again, do 10 more, rest, and then do 10 more. Static exercises such as planks or edge hangs on a board are usually noted in seconds, with each second held counting as one "rep." If you are training limbs individually such as in a lunge or single-arm hang, reps are generally noted in terms of # + #, such as 8+8...which means 8 reps per limb.

If the positions are held for time, reps will be noted by adding "s" or "sec" after the number. A hangboard position done for three sets of 10 seconds per hang would be noted as 3 x 10s.

CIRCUITS

These are groups of several different exercises done in sequence. In a circuit, you perform one set of an exercise before moving on to the next exercise. Once you have completed one set of each

exercise, you start the circuit again. Sessions that feature circuits will usually call for 3-8 circuits of 5-10 exercises. These are typically more endurance/conditioning-based sessions.

SUPERSETS

Supersets are two exercises done in an alternating fashion. In many of our programs, we'll superset a leg exercise and an upper body exercise, with the goal being a shorter, more efficient session. While exercising the legs, the upper body gets some degree of rest, and vice-versa. In a session, this might be noted as:

3 SETS:

5 deadlift

5 bench press

OR:

3X

A1: 5 deadlift

A2: 5 bench press

The A1, A2, etc. notation is indicative of the training group (A, B, C), and the exercise sequence (1,2,3).

TRI-SETS (OR “GIANT SETS”)

Tri-Sets are groups of three exercise performed in series. This can be seen as a “mini-circuit.” In tri-sets, it is best to alternate movement patterns, such as doing a tri-set of upper body, core, and lower body. This is an efficient way to train in the weight room or on the hangboard. Our Integrated Strength sessions are based on this model, and usually follow the hangboard/weight exercise/mobility exercise model. If you need a testosterone boost, you can also call these “giant sets.”

A tri-set might be noted as:

A1: 5 pull-ups

A2: 10 Front Squat

A3: 5 Ab Wheel

3X3 OR 4X5, ETC.

A “number x number” notation is indicative of sets x reps. If an exercise calls for 3x5 you should plan on doing three sets of five reps. Recall that an exercise performed for reps on each side is indicated by the plus (+) sign, so you might see 3 x 8+8 to indicate 3 sets of 8 reps on each side. Reps might also be noted on a set-by-set basis. In this case, you might see 5-3-2 in your session plan, indicating that the first set should be done for 5 reps, the second for 3, and the last for 2. This is usually followed by an explanation of loading parameters.

RM

RM stands for “repetition maximum.” This value is useful in determining the intensity of loading in weight training. For example, your program might call for 3 sets of 3 at 85% of 1RM. In order to effectively use this information, you will need to calculate your 1-rep max for a given movement, then take the percentages. We also use variations on this idea. %MVIC, or the percentage of maximum voluntary isometric contraction, is used in calculating holds of static positions. This would be noted as a percentage of what you could hold for a given amount of time at a given load. “Sec max” is also used. This number, such as “10 sec max” is used in prescribing static holds, as we see in prescriptions on the hangboard.

#:#, SUCH AS 7:13

This notation is used to describe timed intervals. Simply put,

the first number is the work duration and the second is the rest duration. 6 x 7:13 would mean 6 sets of 7 seconds work followed by 13 seconds rest. If the rest is not passive - sitting on the floor or walking around the gym - it is usually noted. For example, in a Rhythm Interval, you have an active rest. The sessions are usually noted as follows: 4 x 30:30, but in the session's description you are instructed to move up and down the board for 30 seconds then rest on a big jug for 30 seconds.

ROUNDS

This is an indication of how many times you would complete a circuit or group of exercises. To use the Rhythm Interval example, you may be instructed to do 3 rounds of 4 x 30:30 with 4 minutes' rest between. With this information, it is easy to figure out just how long your training should take, in the example above, you'd do 4 minutes of intervals in set 1, rest 4 minutes, do 4 minutes of intervals in set 2, rest 4 minutes, and do a final set of 4 minutes...20 minutes total.

SERIES

In endurance training, a series is normally a group of several rounds. This is rarely used notation, but since series and rounds are often used interchangeably, I want to make this distinction clear in the context of this book. Here is an example: An athlete is assigned to do pull-ups, squats, and edge hangs. These are done in a circuit (one set of each in sequence) for 3 rounds. This means the athlete would go through the circuit 3 times. He is assigned 3 series of 3 rounds, though, which means he would end up doing each exercise for 9 total sets, separated by whatever rest is suggested between rounds.

SERIES ONE

Exercise	1
Exercise	2
Exercise	3

Round 1

Exercise	1
Exercise	2
Exercise	3

Round 2

Exercise	1
Exercise	2
Exercise	3

Round 3

REST

SERIES TWO

Exercise	1
Exercise	2
Exercise	3

Round 1

Exercise	1
Exercise	2
Exercise	3

Round 2

Exercise	1
Exercise	2
Exercise	3

Round 3

TECHNICAL FAILURE

The era of bodybuilding ushered in the idea of training to absolute failure. This was relatively safe to do in bodybuilding workouts as the exercises were highly isolated - curls, calf raises, leg extensions and the like. You keep trying to move a load with worse and worse form in search of the elusive “pump.” In the early 2000s high-intensity timed circuits became popular and with the advent of Crossfit so did including compound lifts in these circuits. Training to failure was encouraged and many injuries and serious medical issues were the result.

We encourage athletes who are trying to develop strength or power or skills to train only to the point of technical failure, the point

at which you can no longer do the movement with the same quality or speed as you could at the beginning of the set. There is little value in going past this point except in the pursuit of hypertrophy (bulking up). Since hypertrophy is rarely our goal, we rarely go there.